

Theory 2. Convex sets and functions.

All tasks cost 1 points, so the total sum for all tasks is 10.

Task 1 Suppose that C is a convex set in some linear space. Show that (a) closure $\overline{C}$ and (b) interior $\text{int}(C)$ of the set C are also convex sets.

Task 2 Let C and D be some sets in linear space. Show that:

- (a) $\text{Conv}(C \cup D) = \text{Conv}(\text{Conv}(C) \cup \text{Conv}(D))$.
- (b) $\text{Conv}(C \cap D) \subseteq \text{Conv}(C) \cap \text{Conv}(D)$.

Task 3 Show that the following set is convex: $\{x \in \mathbb{R}^n : x^T P x \leq (c^T x)^2, c^T x \geq 0\}$, where $P \in S_{++}^n$, $c \in \mathbb{R}^n$.

Task 4 By considering operations that preserve convexity, show that the following sets are convex:

- (a) $\{X \in \mathbb{S}^n \mid A^T X + X A + I \preceq 0, \text{tr}(X) \leq 1, X \succeq 0\}$, $A \in \mathbb{R}^{n \times n}$.
- (b) $\{(u, X) \in \mathbb{R}^n \times \mathbb{S}^n \mid u^T X^{-1} u \leq 1, I \preceq X \preceq 2I\}$. *Hint: use the property with Schur complement*

Task 5 Show that $\text{Conv}\{xx^T : x \in \mathbb{R}^n, \|x\|_2 = 1\} = \{A \in S_+^n : \text{tr}(A) = 1\}$.

Task 6 Prove the differential criterion of convexity: a differentiable function f is convex if and only if $f(y) \geq f(x) + \nabla f(x)^T (y - x)$ for all $x, y \in \text{dom} f$.

Task 7 By considering operations that preserve convexity, show that the following functions are convex:

- (a) $f(\mathbf{x}) = \log(\sum_{i=1}^n \exp(\max\{0, x_i\}))$.
- (b) $f(\mathbf{x}) = \inf_{P \in M} (\mathbf{x}^T P^{-1} \mathbf{x} + \text{tr}(P))$, where $M = \{P \in \mathbb{S}^n \mid I \preceq P \preceq 5I\}$. *Hint: use the property with Schur complement and Epigraph criteria of convexity*

Task 8 Let's consider the function $f : \mathbb{R}_{++}^n \rightarrow \mathbb{R}$, $f(x) = \prod_{i=1}^n x_i^{\alpha_i}$, $\alpha_1, \dots, \alpha_n \geq 0$, $\sum_{i=1}^n \alpha_i = 1$. Show that this function is concave.

Task 9 Show that the following function is convex:

$$f(x) = \frac{1}{x_1 - \frac{1}{x_2 - \frac{1}{\ddots - \frac{1}{x_n}}}}.$$

Here the function is defined on a subset of $\mathbb{R}^n$, where every denominator is strictly positive.

Task 10 Prove that continuous function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if and only if

$$f\left(\frac{x+y}{2}\right) \leq \frac{f(x) + f(y)}{2}, \quad \forall x, y \in \mathbb{R}^n.$$